Understanding Git by practice: day-to-day workout

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`Git` is a version control system that makes it easier to save, manage the evolution of and share a software project. It has become an essential tool in the field of *data science*. This chapter presents a few concepts that will be put into practice in the next one.

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The introductory chapter of this part outlined what is at stake, which is summarised in a dedicated course I gave with Romain Avouac (in French).

Scroll down to read the *slides* below or click here to display them full screen (the slides are in French). This chapter is your opportunity to take your first steps with `Git`. The next chapter is devoted to collaborative work. To make learning easier, this chapter suggests using the `Git` extension of `VSCode` or `JupyterLab`. `VSCode` probably offers, at the moment, the most complete extension. Some parts of this tutorial require the command line.

It is perfectly possible to complete this tutorial entirely from the command line. However, for someone new to `Git`, a graphical interface can be a valuable aid to understanding and adopting `Git`. Once you are comfortable with `Git`, you can easily do without graphical interfaces for daily routines and only use them for certain operations where they prove very handy (notably comparing two files before having to merge them).

To understand the analogies with hand-made versioning, let us recall how it works with Figure 1

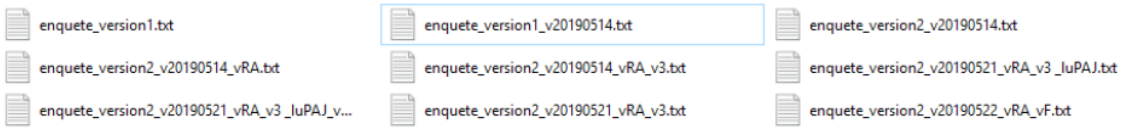


Figure 1: Handmade version control

! Important

It is strongly recommended to use `VSCode` to learn `Git`. Its extension is very well done, much better than the `Jupyter` one.

`Colab` does not natively come with a `Git` extension. Automatic backups to `Github` are possible but this is not a practice to encourage. Even worse, `Colab` will rather offer an integration with `Drive`, another `Google` product. The *notebook* will indeed be versioned since `Drive` keeps version backups, but it is not a technology designed for code backups; it will not bring the benefits of `Git` that will be discussed later.

ENSAE students, and more generally everyone who can benefit from the SSPCloud infrastructure[1], have access to `Python` development environments with `Git` preinstalled and accessible through interfaces connected to IDEs. This *notebook* can be launched on this infrastructure using these buttons

If you are not eligible for the `SSPCloud`, the path to a ready-to-use environment for `Git` and `Python` is more winding. It is recommended to download and install `VSCode` and to add, at the very least, the `Python` extensions and `GitLens`. You can of course customise your development environment much further, but these are the minimal building blocks for a functional and ergonomic local environment.

1 Before starting: create a `Github` account and a working copy

The first step takes place on `Github` and consists in creating an account on this platform.

Exercise 1a: Create a Github account

It is important to follow the instructions step by step, every step matters

1. If you do not already have one, create an account on github.com

Log in at https://github.com > + (at the top of the page) > New repository > Fill in the "Repository name" > Tick "private" > "Create repository"

> 👉 **Repository**: file tree whose history we want to keep in a common place.

Exercise 2: create a repository on Github

- Create a repository following the instructions below.
  - **Create this repository as private**, this will allow us to activate our token in exercise 3. You can make it public after exercise 3, it is up to you.

[1] To find out whether you are eligible for the SSPCloud, you can click on this link and check the list of authorised domains.

- **Create this repository with a `README.md`** by ticking the `Add a README file` box
- **Add a `.gitignore`** by selecting the `Python` template

2 Some `Git` basics

2.1 Remote version, local version

With the previous exercise, we created a first repository. It is a centralised repository that will serve as the source of truth for our project and through which the contributors to a project interact. But we have not yet discussed how to make it evolve; for that, we need to create working copies.

`Git` is a decentralised version control system[2]. This means that contributors edit files in their favourite editor and then submit them to update the source of truth, the remote repository.

> 👉 **Version control**: practice of tracking and managing the changes made to a software project.

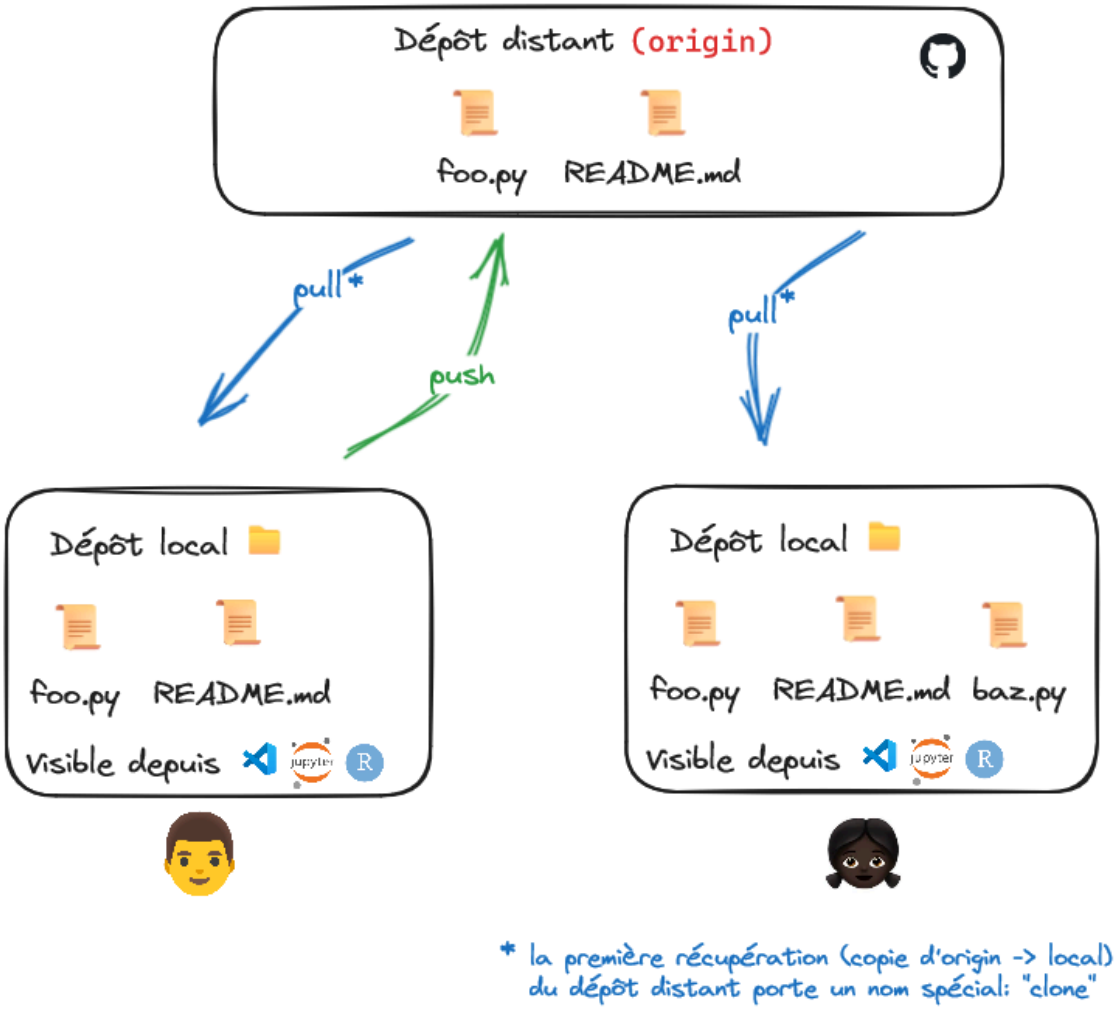


Figure 2: The decentralised principle of `Git`

[2]More precisely, `Git` is a decentralised and asynchronous version control system. This means that, besides the fact that files are edited on local copies, there is no need to be continuously connected to the remote repository. You can make changes and submit them later.

We will come back to Figure 2 in more detail later, in particular to the many technical terms shown on it. But understanding this fundamental distinction between remote and local repositories was important in order to get started. A platform that stores remote repositories is called a **forge**. In this course, we will present `Github` but there are others, notably `Gitlab`.

> 👉 **Forge**: collaborative management system for text or code. `Github` and `Gitlab` are the two best-known forges but there are many others.

We created our remote repository in the previous exercise. How do we create a working version? This operation is called cloning (`git clone`). The goal of the next exercise is to do this, but it first requires understanding the concept of authentication before we can start.

> 👉 `clone`: retrieving a remote repository and its history by creating a local copy

Although it is possible to use `Git` offline, that is, pure local version control without a remote repository, this is a rare use with limited interest. The value of `Git` is that it offers a robust and efficient way of interacting with a remote repository, thereby facilitating collaboration, whether in a team or alone.

> 💡 Why Github?
>
> For these exercises, we suggest using `Github`, the most visible forge. The advantage of `Github` over its main competitor, `Gitlab`, is that it is more visible, because it is better indexed by `Google` and, partly for historical reasons, gathers more `Python` and `R` developers (which matters in fields like code where network externalities play a role).
>
> Being familiar with the `Gitlab` environment is still useful because many internal software forges are built on the *open-source* features (the graphical interface included) of `Gitlab`. It is therefore very useful to master the core features of these two interfaces, which are in fact almost identical. Conveniently, this is the subject of this chapter and the next.

2.1.a Authenticating to `Github` with a token: principle

> 👉 **Authentication**: process allowing a computer system to make sure of the identity of whoever wants to perform an action.

`Git` is a decentralised version control system: code is modified by each person on their own workstation, then brought into line with the collective version available on the remote repository whenever the contributor decides.

The forge therefore needs to know the identity of each contributor, in order to determine who is the author of a change made to the code stored in the remote repository. For `Github` to recognise a user who proposes changes, they must authenticate (a remote repository, even a public one, cannot be modified by just anyone). Authentication thus consists in providing something that only you and the forge are supposed to know: a password, a complicated key, an access token…

More precisely, there are two ways to make your identity known to `Github`:

- an **HTTPS authentication** (described here): authentication is done with a login and password or with a *token* (a complicated password automatically generated by `Github` and known only to the holder of the `Github` account);
- an **SSH authentication**: authentication is done with an encrypted key available on the workstation and known to `GitHub` or `GitLab`. Once configured, this method no longer requires you to state your identity: the fingerprint that the key constitutes is enough to recognise a user. This is not the method we will apply here[3].

[3] The collaborative `utilitR` documentation (in French) explains why the HTTPS method should be preferred over the SSH method.

ℹ A note on two-factor authentication

> 👉 **Two-factor authentication** (2FA): security practice that authorises authentication only after two distinct proofs of identity have been presented to an authentication mechanism. For instance, in a banking app, providing a customer number and a code sent by SMS

Since August 2021, `Github` no longer allows password authentication when interacting (`pull`/`push`) with a remote repository (reasons here). You must use a *token* (access token), which has the advantage of being revocable (you can delete a token at any time if, for example, you suspect it was leaked by mistake) and of having limited rights (the token allows certain standard operations but does not allow certain critical operations such as deleting a repository).

GitHub will progressively require all GitHub users to enable one or more forms of two-factor authentication (2FA). For more information on the 2FA rollout, see this blog post. In practice, this means you will have to choose between:

- Providing your mobile phone number to validate some logins with a code you will receive by SMS;
- Installing an authentication app on your phone (e.g. Microsoft Authenticator) that will generate a QR code you can scan from GitHub, which does not require you to provide your phone number
- Using a USB security key

To choose between these options, go to *Settings > Password and authentication > Enable two-factor authentication.*

2.1.b Creating a token

> 👉 **Token authentication** is a form of authentication that allows a user to access an online service, an application, or a website without having to enter their credentials again. Authentication tokens work like an entry ticket with limited validity: they grant continuous access for as long as they are valid. As soon as the user logs out or leaves the application, the token is invalidated.

The official documentation contains a number of screenshots explaining how to proceed. Keeping this documentation open in case of doubt, together with the instructions of the next exercise, we can create an authentication token.

Exercise 3: Create a token

Follow the official documentation, giving the token only the `repo` rights[4].

To sum up, the steps should be as follows:

Settings (account) > Developers Settings > Personal Access Token > Tokens (classic) > Generate a new token (classic) > "MyToken" > Expiration "90 days" > tick only "repo" > Generate token > Copy it

⚠ **Keep the page open**, the *token* only appears once and we have not yet made the effort to store it somewhere durable. That will be the subject of the next exercise. Do not worry, though, if you lost the *token* before you could save it: you can generate a new one by following the procedure above again.

We have created a *token* and, as indicated on the `Github` page or in the exercise instructions, it is not permanent. We therefore need to find a way to keep it somewhere. We will suggest several solutions.

[4] As you can see, there are many different levels of rights. Your password actually has all these rights, which illustrates how dangerous it is if it is discovered, whether through an accidental disclosure or through someone malicious *hacking* it. Using a *token* makes your repository much safer since it cannot be deleted if the *token* has limited rights. Moreover, if the main branch is protected, which is the default behaviour of `Github`, it will not be possible to destroy the repository history without a password.

Writing it in a text file created with a notepad is not one of them; on the contrary, it is a very bad practice.

> ! Important
>
> It is important never to store a *token*, let alone your password, in a project. It is possible to store a password or *token* in a secure and durable way with the `Git` *credential helper*. It is presented later on.
>
> If it is not possible to use the `Git` *credential helper*, a password or *token* can be stored securely in a password manager such as Keepass.
>
> Never store a `Github` token, or worse a password, in an unencrypted text file. Password managers (such as Keepass, recommended by ANSSI, the French national cybersecurity agency) are easy to use and only keep on the computer a hashed version of the password that can only be decrypted with a password known to you alone.

> Exercise 3bis: store your token (for SSPCloud users)
>
> The `SSPCloud` offers a token storage service that can then easily be used to authenticate to `Github` when using `VSCode` or `Jupyter`.
>
> - Copy the token displayed on the `Github` page you opened earlier. Do not select the characters by hand, use the dedicated copy button ()
> - Click on the "My Account" section of the `SSPCloud` (*Mon Compte* if the interface is in French). Go to the `Git` tab and paste the value you copied.
>
> You can stop at this point, we will use this token in the next exercise.

> Exercise 3ter: store your token (for people without SSPCloud access)
>
> The recommended solution is to store your token in a password manager such as Keepass (recommended by ANSSI). It is software that stores, in encrypted form, the passwords kept in it, following the logic of a digital safe.
>
> Beyond the benefit of storing a Github token for this course, such software is very handy day to day and secures access to sensitive digital services. It also includes strong password generators that reduce the risk of digital identity theft by making techniques like brute-force attacks almost impossible.

Now that we have stored our token in a secure place, we can move on to the next step, which is to retrieve our remote repository into a working copy, an operation that will lead us to actually use the token we set aside for now.

> Exercise 4a: create a service and understand the principle of cloning and authentication (SSPCloud users)
>
> 1. On the SSPCloud, go to the My Services page (*Mes Services* if the interface is in French).
> 2. Click on ➕ `New service` (*Nouveau service*) and choose a `vscode-python` service (do not pick another one).
> 3. Keep the default parameters and launch the service.
> 4. Once the service is ready, click on the button "Click to copy the service password" (*Cliquez pour copier le mot de passe du service*). This will store the service password (randomly generated, it has nothing to do with your general `SSPCloud` password) in the clipboard. This password is also visible in plain text in the part that is blacked out in Figure 3.

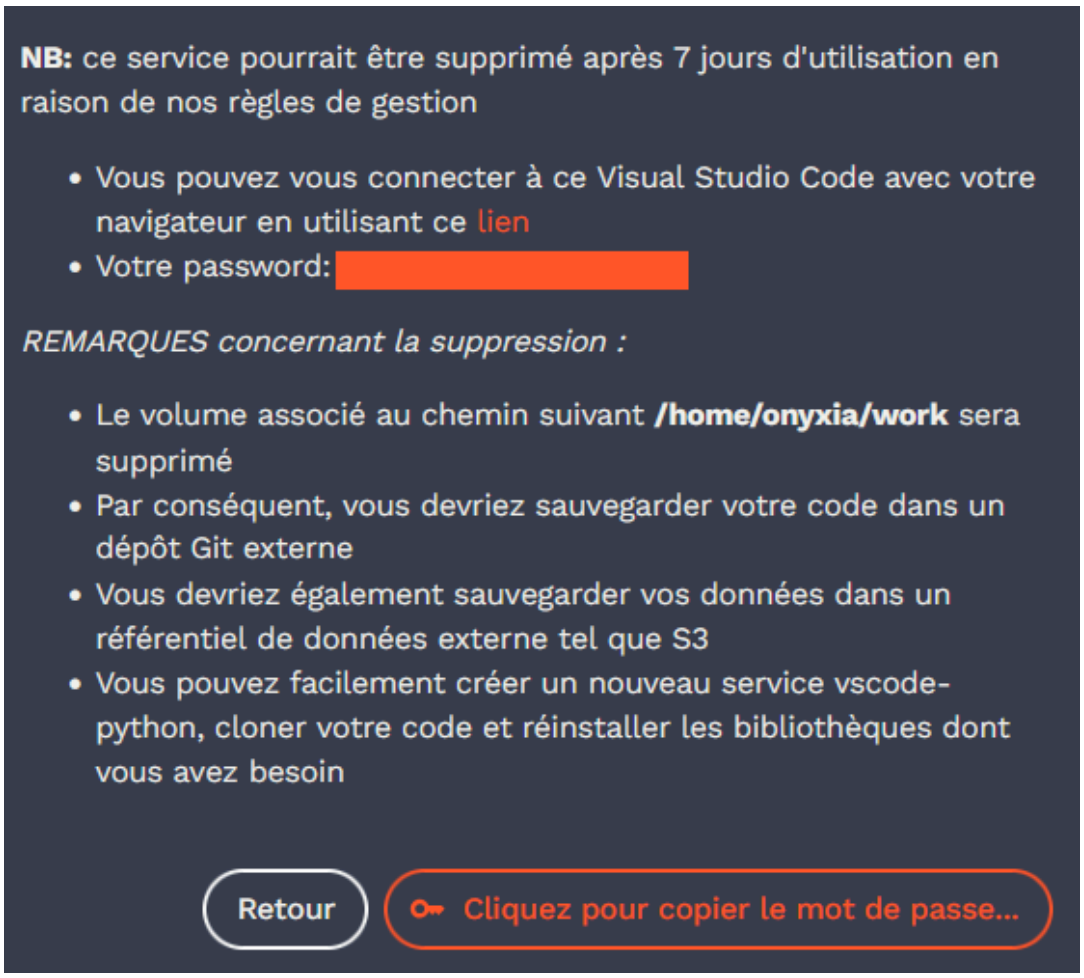


Figure 3

5. Paste this password into the `Password` field that appears when you open the service.
6. In another tab, retrieve, from the home page of your repository, the URL of the remote repository, which is available by clicking on the green `<> Code` button on the right. The URL has the following form

`https://github.com/<username>/<reponame>.git`

7. Open the terminal (`☰ > Terminal > New Terminal`) and start typing

```
git clone # paste your url of the form https://github.com/<username>/<reponame>.git
```

to paste your URL after it; if CTRL+V is blocked by the browser, you can use SHIFT+Insert. Press Enter.

8. A page will open: *"The extension 'GitHub' wants to sign in using GitHub"*. Refuse by clicking on *"Cancel"* (the optional questions show what happens when you accept: you switch to another authentication mode).
9. In the window at the top, type your username first. Then, when it asks for your password, **paste your *token*, not your `Github` password** (if you still have the `Github` page open, copy it from there, otherwise go back to the My account page of the `SSPCloud`)
10. Observe the update of the file explorer in `VSCode`: your `README` and your `.gitignore`, which are visible on `Github`, should now be there.

11. Type `cd my-little-project` assuming the folder of your repository is called `my-little-project`. Then type `git remote -v`, a command that asks `Git` where `origin`, your remote repository, points. The answer should be the URL you entered previously

This was a necessary illustration to understand the principle of authentication. The next exercise (4b) will suggest a more direct way of working, which is useful to know because it will save you from having to authenticate at each interaction with the remote repository.

Optionally, to understand the difference with the delegated authentication offered by VSCode, you can follow these optional instructions:

1. Still in the same VSCode, open a new terminal (`☰ > Terminal > New Terminal`)
2. Type `git clone https://github.com/<username>/<reponame>.git repo-bis`, replacing `https://github.com/<username>/<reponame>.git` with the URL of your repository. This will clone your repository into the `repo-bis` folder once you are actually authenticated.
3. This time, accept the delegated authentication offered by VSCode. It is a two-factor authentication:
    - The first authentication factor is the code that `Github` asks you to copy and to enter in the page that `VSCode` wants to open (you have to accept to copy it and to open the page). Paste this 8-character code, validate and accept the rights requested by the application.
    - The second factor is the code from your authentication app (for example `Google Authenticator`, or the one you receive by SMS). Enter it and validate, and the clone should start

This exercise has just illustrated the principle of authentication and the way `VSCode` can attest to your identity thanks to a *token* or to two-factor authentication. The next exercise suggests a token authentication method that is a bit more practical than the one we implemented 🤘.

Exercise 4b: create a service and understand the principle of cloning and authentication (SSPCloud users)

This approach shows how the SSPCloud injects the *token* and the repository you want to clone when a VSCode service is created.

1. On the SSPCloud, go to the My Services page. You can delete the existing service, it is no longer needed.
2. Click on `➕ New service` and choose a `vscode-python` service (do not pick another one).

- In another tab, retrieve, from the home page of your repository, the URL of the remote repository, which is available by clicking on the green `<> Code` button on the right. The URL has the following form

`https://github.com/<username>/<reponame>.git`

- Unfold the `Vscode-python configuration` menu (*Configuration Vscode-python*) and look for the `Git` tab
- In it, you should see your *token* pre-injected in the form. Do not change it.
- In another tab, retrieve, from the home page of your repository, the URL of the remote repository, which is available by clicking on the green `<> Code` button on the right. The URL has the following form

`https://github.com/<username>/<reponame>.git`

You can use the icon on the right to copy the URL.

- Paste it into the `Repository` field of the service creation form on the `SSPCloud`. Launch the service and wait for it to be created (about twenty seconds).
- The clone of the remote repository should be visible in the file tree.

- Open the terminal (`☰ > Terminal > New Terminal`) and type `git remote -v`, a command that asks `Git` where `origin`, your remote repository, points. The answer takes the form:

```
https://ghp_XXXX@github.com/username/repository.git
```

which differs from the URL you entered in the `Git` tab. As you can see with this method, the token is in plain text. This is why tokens are used rather than passwords: if they are revealed, they can always be revoked, which avoids problems

Alternative exercise 4: when you do not have an SSPCloud account

The procedure is very similar. In practice, the only difference is that there is no need to create a new service since a `VSCode` installation already exists.

1. In your browser, retrieve, from the home page of your repository, the URL of the remote repository, which is available by clicking on the green `<> Code` button on the right. The URL has the following form

`https://github.com/<username>/<reponame>.git`

7. Open the terminal (`Terminal > New Terminal`) and start typing

```
git clone
```

and paste the value you copied earlier. Do not press Enter yet. With the arrow keys, move between `https://` and `github.com`. Retrieve your `Github` token from your browser or from `Keepass`. Paste it then add `@`. This should give

```
https://ghp_XXXX@github.com/username/repository.git
```

Which, all in all, gives

```
git clone https://ghp_XXXX@github.com/username/repository.git
```

- The clone of the remote repository should be visible in the file tree.

As you can see with this method, the token is in plain text. This is why tokens are used rather than passwords: if they are revealed, they can always be revoked, which avoids problems

2.2 The *staging area*

> 👉 ***Staging area***: `Git`'s waiting area before new changes are validated into a file's history.

In a world without `Git`, you write code, save your script and sometimes decide that this version is worth treating as one you can start over from. With `Git` it is the same thing, except that the principle is formalised more cleanly.

The first conceptual level is the index of changes. These are the changes awaiting validation, hence the name *staging area* in the first part of Figure 4.

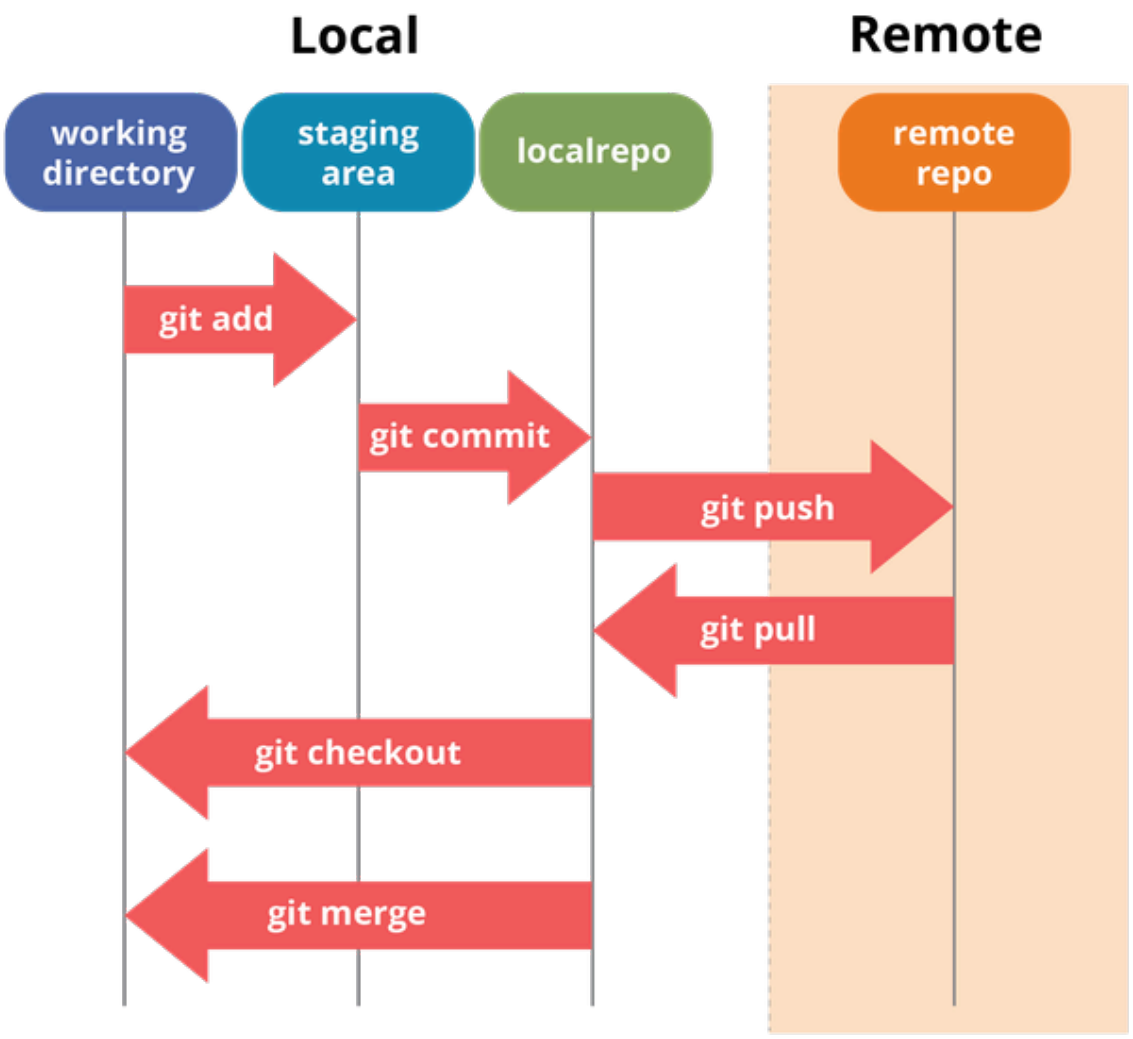


Figure 4: The whole `Git` workout

In principle, when you edit scripts or *notebooks*, you save them regularly. The next level of commitment is to set aside a particular version of them, which by hand (see Figure 1) we would do by duplicating the file. This means putting one or more changes in the waiting list of changes to be validated. This operation is called `git add` and is the subject of the next exercise.

Exercise 5: Stage changes

1. Create a 📁 `scripts` folder in the folder of your repository. In `VSCode`, you can use the appropriate icons.
2. Create the files `script1.py` and `script2.py` in it, each containing a few `Python` commands of your choice (the content of these files does not matter).
3. Go to the `Git` extension of `VSCode`. You should find a panel that looks like this

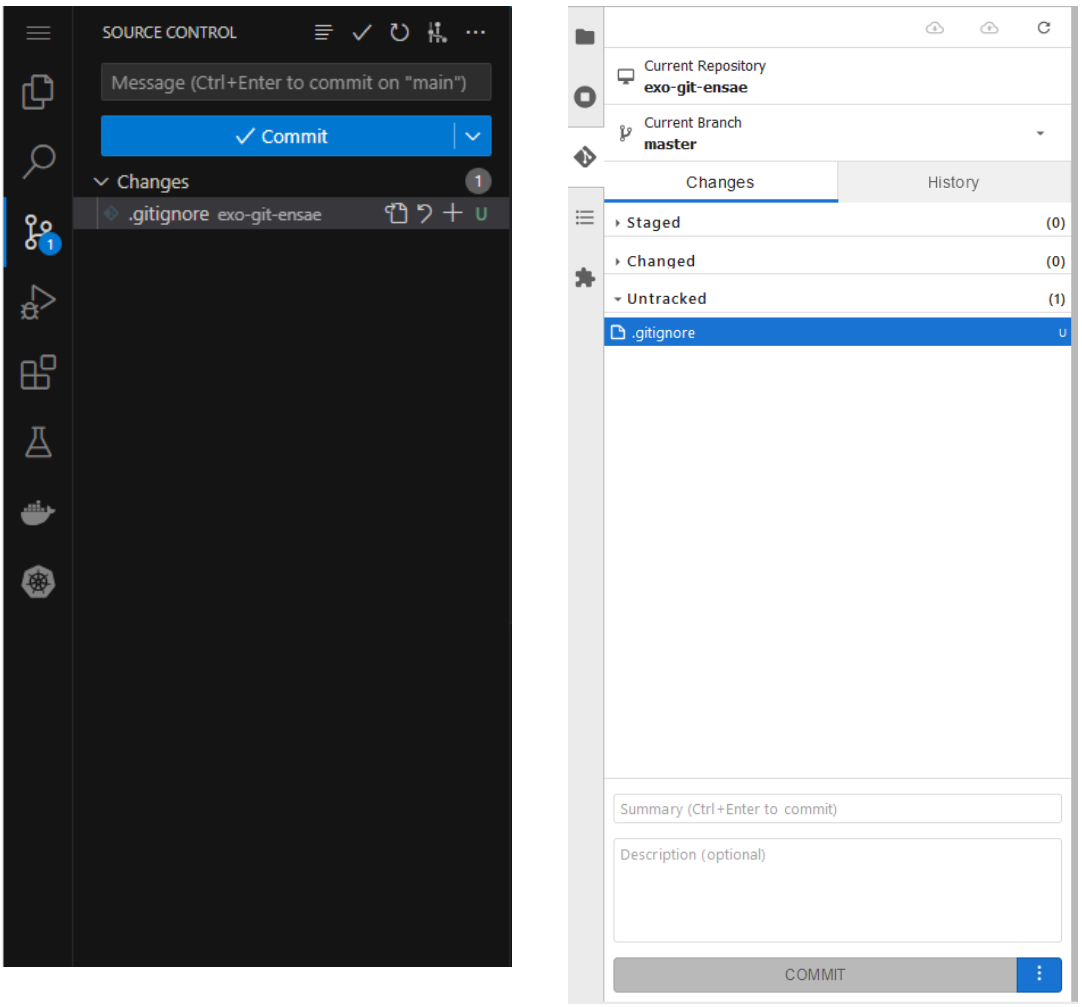


Figure 5: Git graphical interface in VSCode (left) and Jupyter (right)

On the command line, this is the equivalent of

```
git status
```

2. In `VSCode`, a `+` button appears to the right of the name of the files `script1.py` and `script2.py`. In `Jupyter`, when you hover your mouse over the names of the files `script1.py` and `script2.py`, you should see a `+` appear. Click on it.

If you had preferred the command line, what you did is equivalent to:

```
git add scripts/script1.py
git add scripts/script2.py
```

3. Observe the change in the file's status after clicking on `+`. It is now in the `Staged` section.

Basically, you have just told `Git` that you are going to make a change to the file public, but you have not done it yet (`Staged` is a waiting list).

If you were on the command line, you would get this result after a `git status`

```
On branch main

No commits yet
```

```
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
```

The new changes (in this case the creation of the file and the validation of its current content) are not yet archived. For that, we will need to make a `commit` (we commit ourselves by making a change public).

4. Modify the content of an existing line in the `README.md` and go through the same staging routine.
5. Let us look at the changes we are about to validate. To do so, hover over the name of the file `README.md` and click on it. A page opens and puts the previous version side by side with the new one, with additions in green and deletions in red. We will find this visualisation again with the `Github` interface, later.

If you do the same for `scripts/script1.py`, since the file did not exist, we normally only have additions.

It is also possible to do this with the command line but it is much less convenient. The command to call is `git diff` and you need to use the `cached` option to tell it to inspect the files for which we have not yet made a `commit`. Changes will appear in green and deletions in red but, this time, the results are not shown side by side, which is much less convenient.

```
git diff --cached
```

2.3 The *commit*

At this stage, we have configured `Git` so that it can authenticate automatically and we have cloned the repository to get a local working copy. We made changes and put them in the waiting queue of our version control system. All that remains is to finalise the work.

As a reminder, Figure 4 illustrated the routine for making the history of your project evolve. After adding to the queue (the *staging area*), you validate the changes with a *commit*. As its name suggests, it is a change proposal to which we, in a way, commit ourselves.

> 👉 ***Commit***: Record of the evolution of a file. It is the fundamental unit of time in `Git`.

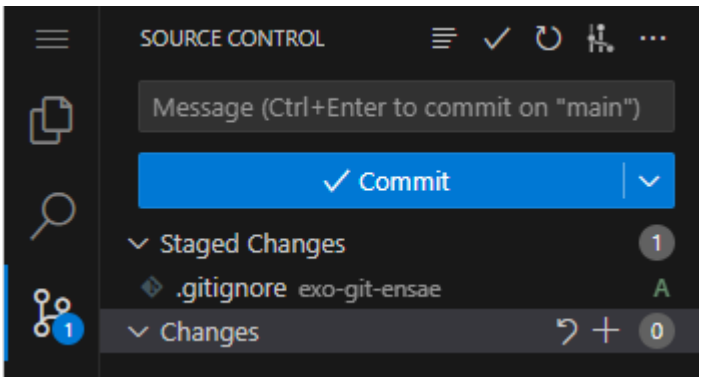
A *commit* has a title and optionally a description. To this information, `Git` will automatically add a few additional elements, notably the author of the commit (to identify the person who proposed this change) and the timestamp (to identify when this change was proposed). This information makes it possible to identify the `commit` uniquely, and a unique random identifier (a SHA number) is added to it, which makes it possible to refer to it without ambiguity.

The title is important because it is, for a human, the entry point into the history of a repository (see for example the history of this course's repository). Vague titles (*File update*, *Update*...) should be banned because they require needless effort to understand which files were modified.

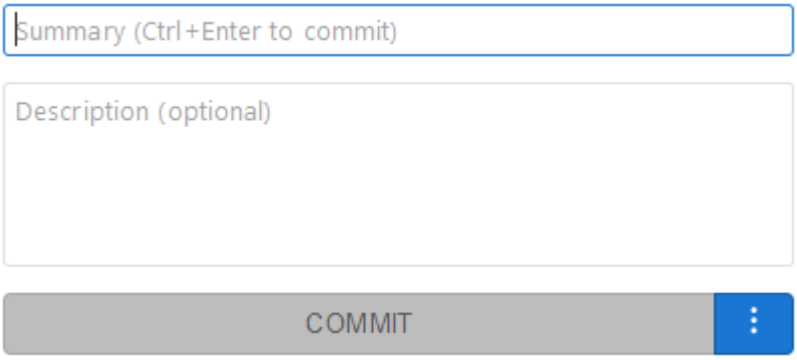
Do not forget that your first collaborator is your *future self* who, in a few weeks, will not remember what the *Update* commit of 12 January was about and how it differs from the *Update* of 13 March.

Exercise 5: first commit (at last!)

In `VSCode`, on the contrary, you have to look at the top.



On the `Jupyter` interface In `Jupyter`, conversely, everything happens in the lower part of the graphical interface.



1️⃣ Enter the title `Create the first scripts` 🎉 and add the description `Files in the scripts folder`. In `VSCode`, the title of the commit corresponds to the first line of the commit message; the following lines, after a line break, correspond to the description.

2️⃣ Click on `Commit`. The file has disappeared from the list, this is normal: it no longer has any change to validate. To find it again in the list of `Changed` files, you will have to modify it again

3️⃣ You should see the lifeline of your project grow longer. You can click on the commit to find the changes you validated

i Note

If you were using the command line, the equivalent way of doing this would be

```
git commit -m "Initial commit" -m "Create the first files 🎉"
```

The `m` option creates a message, which will be available to all contributors of the project. On the command line, this is not always very convenient. Graphical interfaces allow for more developed messages (good practice is to write a commit message like a succinct email: a title and a little explanation, if needed).

2.4 The `.gitignore` file

When using `Git`, there are files you do not want to share or whose changes you do not want to track (typically large databases).

It is the `.gitignore` file that manages the files excluded from version control. When creating the project on `GitHub`, we asked for a `.gitignore` file to be created, located at the root of the project. It specifies all the files that will always be excluded from `Git`'s indexing.

> 👉 `.gitignore`: file storing a set of rules for files that must not be tracked by `Git`.

Exercise 6: the .gitignore file

1️⃣ Open the `.gitignore` file and look at some of the rules written in it

Display this file when using `Jupyter` By default, the `.gitignore` file is not displayed because `.*` files are configuration files. You need to enable an option to display it. At the very top of `Jupyter`, click on `View → Show Hidden Files`

2️⃣ Create a `data` folder at the root of the project and, inside it, create a file `data/raw.csv` with any line of data. Look at the `Git` tab and notice that the file appears among the changes to be validated. **Do not add this file to the *staging area* nor commit it**.

3️⃣ Add the line `data/` to the `.gitignore` (it is an editable text file), do not forget to save. Go back to the `Git` tab and observe the change. Do you understand what is happening?

4️⃣ Create a file `raw2.csv` at the root and a file `data/read_data.py` with a line of `Python` code (the content of the file does not matter).

5️⃣ Go back to the `Git` tab and observe the change in your repository.

The `.gitignore` should not be neglected.

`Git` is a version control system designed for text files, not data files. Unlike text files, where `Git` can track changes line by line, which makes it a useful version manager, data files are too complex for this fine-grained tracking, so every change to these files, which are often heavy (several megabytes), amounts to duplicating the file in `Git`'s memory (more details on the alternatives in Tip 1).

Besides this technical reason for wanting to exclude data files from version control, do not forget that the data used by *data scientists* is often confidential, collected for well-defined purposes or of strategic interest that disclosure to competitors could weaken. Sharing data with the whole world can be very costly under the GDPR, whose fines can reach 4% of worldwide turnover.

This `.gitignore` file thus acts as a safeguard. It is useful to be conservative with it, even if it means allowing exceptions, case by case and consciously. For this reason, it is recommended to add many classic data extensions to it: `*.xlsx`, `*.csv`, `*.parquet`...

Tip 1: If data does not go on Github, where should it be stored?

It is more relevant to store data in *ad hoc* environments. For this, the state of the art in *cloud* technologies is to use a specialised storage system that follows a protocol called `S3`. It was originally developed by Amazon for its commercial AWS *cloud* before being made *open source*. This makes it possible to have *cloud* infrastructures that use this technology independently of Amazon.

`Github` admittedly seems convenient for storing data. But there is no such thing as a free lunch. This first has an environmental cost: `Git`, and *a fortiori* `Github`, are not database storage systems, so they do not optimise the storage of data. We should avoid increasing the carbon footprint of digital technology, already growing [1], any more than necessary through digital waste. Then, this choice, originally convenient, often proves costly in the long run. If the data archived with `Git` changes frequently, the unavoidable duplication it implies will make the volume of the repository explode. Beyond a certain threshold, you will have to switch to complex solutions such as `Git large file storage (LFS)`, which is clearly avoidable with discipline. In contrast, with storage systems designed for data, such as the `S3` protocol, storing and processing data are optimised. This is what distinguishes data storage systems like `S3` from other storage systems designed for files, regardless of their format, like `Drive`, which is not suited to the needs of *data scientists*.

Storing data is expensive for *cloud* providers and it is no surprise that they rarely offer free plans for large volumes. `SSPCloud` users benefit from a storage system attached to the platform and following the S3 protocol. More details in the dedicated chapter and in the ENSAE 3rd-year course Putting *data science* projects into production (in French).

3 First interactions with `Github` from your working copy

So far, after cloning the repository, we have worked only on our local copy. We have not tried to interact again with `Github`.

It is perfectly possible to do version control without setting up a remote repository. However,

- it is dangerous since the remote repository acts as a backup of a project. Without a remote repository, you can lose everything if there is a problem with the local working copy;
- it means choosing to be less efficient because, as we will show, the features of the `Github` and `Gitlab` platforms are also very beneficial when working alone.

Interacting with the remote repository consists in retrieving what is new on it or sending our latest changes to this central repository. In practice, we can therefore summarise 95% of daily `Git` practice in three words:

- `commit`: I validate the changes I made locally with a message explaining them
- `pull`: I retrieve the latest version of the code from the remote repository (an operation called `fetch`) and merge it with my changes, if I made any (an operation called `merge`)
- `push`: I send my validated changes to the remote repository

We have already seen the first one, we will now see the two new terms. Before that, we can come back to a command already mentioned, `git remote -v`. The remote repository is called a `remote` in Git language. The `-v` option (*verbose*) lists the remote repository(ies).

By opening a terminal in `VSCode` and typing `git remote -v` in it, you should get a result similar to this one:

```
origin  https://<PAT>@github.com/<username>/<projectname>.git (fetch)
origin  https://<PAT>@github.com/<username>/<projectname>.git (push)
```

Several pieces of information are interesting in this result. First, we find the url we gave to `Git` during the cloning operation. Then, we notice a term `origin`. It is an alias for the url that follows. It saves us from having to type the whole url every time, which can be tedious and error-prone.

`fetch` and `push` are there to tell us that we retrieve (`fetch`) changes from `origin` but also send (`push`) changes to it. Generally, the urls of these two repositories are the same but it can happen, when contributing to open-source projects that we did not create, that they differ. This will be explained in the next chapter when we tackle the subject of collaboration.

3.1 Sending changes to the remote repository (`push`)

Exercise 7: Interact with Github

We now need to send the files to the remote repository.

1️⃣ If you have not already done so, add the rule `*.csv` to your `.gitignore`. With the graphical interface, add all pending files to the index (`git add`) and make a *commit* (`git commit`). 2️⃣ Click on the Sync changes button

The equivalent of these two steps on the command line would be:

1. This means: *"git sends (`push`) my changes on the `main` branch (the branch we worked on, we will come back to it) to my repository (alias `origin`)"*

3.2 Retrieving changes from the remote repository (`pull`)

The second way of interacting with the repository is to retrieve results available online onto your working copy. This is called `pull`.

For now, you are alone on the repository. There is therefore no partner to modify a file in the remote repository. We will simulate this case by using the `Github` graphical interface to modify files. We will bring the results back locally in a second step.

Exercise 8: Bring changes back locally

1️⃣ Go to your repository from the https://github.com interface

- Go to the `README.md` file and click on the `Edit this file` button, which looks like a pencil icon.

2️⃣ Change the title of the `README.md`. Skip a line at the end of your file and enter the text you want, without punctuation. For example,

```
the oak one day said to the reed
```

3️⃣ Click on the `Preview` tab to see the text formatted in `Markdown`

4️⃣ Write a title and an additional message to make the `commit`. Keep the default option `Commit directly to the main branch`

5️⃣ Edit the `README` again by clicking on the pencil just above the display of the `README` content.

Add a second sentence and fix the punctuation of the first one. Write a commit message and validate.

```
The Oak one day said to the Reed:
You have good reason to accuse Nature
```

6️⃣ Above the file tree, you should see the title of the last commit. You can click on it to see the change you made.

7️⃣ The results are on the remote repository but are not in your working folder in `Jupyter` or `VSCode`. You need to re-synchronise your local copy with the remote repository:

- In `VSCode`, simply click on `... > Pull` next to the button that lets you view the Git graph.
- With the `Jupyter` interface, if possible, simply press the small down arrow, the one that now has the orange badge.
- If this arrow is not available or if you work in another environment, you can use the command line and type

1. This means: *"git retrieves (`pull`) the changes on the `main` branch to my repository (alias `origin`)"*

8️⃣ Look at the commit history again. Click on the last commit and display the changes on the file. You can notice how fine-grained version control is: `Git` detects, within the first line of your text, that you added capital letters or punctuation.

The `pull` operation allows:

1. Your local system to check for changes on the remote repository that you have not made (this operation is called `fetch`)
2. To merge them if there is no version conflict or if the version conflicts can be merged automatically (two modifications of a file that do not affect the same location).

4 Branches

So far, we have made our history evolve linearly, without taking any risk. This is already very reassuring since we can always recover a past state of the code. But what should we do when we want to make substantial changes, involving several intermediate steps that are each meant to be a *commit*, without the risk of destabilising the version we were, until now, happy with?

`Git` offers an extremely convenient system for this: branches. A branch is a parallel version of the project that coexists with the main version, on `main`. This means that with `Git`, the same file can coexist in several different versions, which is fertile ground for experimentation. One version is active (the active branch) but the others remain available and can be activated if needed.

You can see the history of `Git` as a refined tree. The `main` branch is the trunk. The other branches start from this trunk and diverge. The trunk may very well evolve in parallel to these branches. The `Git` tree is nevertheless special, somewhat gnarled: branches can come back and merge with the trunk. We will see this in a more illustrated way in the next exercise and in the next chapter since it is one of the foundations of collaborative work with `Git`.

Here are a few examples where branches are used for significant work:

- you work alone on a task that will take you several hours or days of work (you must not push unfinished work to `main`);
- you are working on a new feature and you would like to collect the opinion of your collaborators before modifying `main`;
- you are not sure you will succeed in your changes on the first attempt and prefer to run tests in parallel.

It is common, in a collaborative setting, to use branches lavishly. As we will see in the next chapter, this can indeed save precious time if the project organisation is poorly defined. However, branches are not a magic cure because managing them requires rigour and can lead to disorganisation without it. `Git` is a technical solution that smooths organisation but, when roles are poorly defined, it will quickly reveal the problems without offering the solution unless you put some thought into how work is organised.

Caution

Branches are not personal: just because you created a branch does not mean that one of the people collaborating with you on the `Git` project will not be able to modify it.

Using branches is already an advanced feature of `Git`. It is a wonderful technical tool but it does not solve organisational problems. On the contrary, with `Git`, they will show up much faster. `Git` is really the minimal foundation of good practices, a tool that will push you to always work better.

Among the practices that are technically possible but not recommended, you must **absolutely** ban the use of `push force`, which can destabilise collaborators' local copies. If a `push force` is necessary, it means there is a problem in the branch, which must be identified and fixed **without** a `push force`.



Branches are generally associated with the *issues* system on `Github`. *Issues* are not native `Git` features but are provided by `Github`, which aims to simplify project tracking and feedback from other collaborators or users of a project.

Issues can be seen as a discussion system where people interested in the project can exchange. The advantage of using this through `Github` rather than an email loop is that it centralises, in the same place as the code, the exchanges and documentation useful for its evolution. The purpose of *issues* is very broad: they can be used to discuss new features that will eventually not be implemented, report bugs, share out the work, etc. Intensive use of *issues*, with appropriate labels, can even make project management tools like `Trello` unnecessary.

The next exercise aims to illustrate the principle of branches by making up an example of a request going through an *issue* and proposing a new feature in the project.

Exercise 9: Create a new branch and merge it into main

1️⃣ Open an *issue* on `Github` (see the explanations above on the principle of *issues*). Point out that it would be nice to add a cat emoji to the `README`. On the right-hand side, click on the small cog next to `Label` and click on `Edit Labels`. Create a `Markdown` label. Normally, the label has been added.

2️⃣ Go back to your local repository. You will create a branch named `issue-1`

How to do it in `VSCode`?

In VSCode, click on `... > Branch > Create Branch` and enter the name `issue-1`. How to do it in `Jupyter`?

With the `JupyterLab` graphical interface, click on `Current Branch - Main` then on the `New Branch` button. Enter `issue-1` as the branch name (the branch must be created from `main`, which is normally the default choice) and click on `Create Branch`.

How to do it on the command line?

If you are not using the graphical interface but the command line, the equivalent way of doing it is

1. The `checkout` command is a Swiss Army knife of branch management in `Git`. It lets you switch from one branch to another, but also create branches, etc.

3️⃣ Open `README.md` and add a cat emoji (🐱) after the title. Make a commit by redoing the steps seen in the previous exercises. Do not forget, it is done in two steps:

1. Add the changes to the index by moving the `README` file to the `Staged` section
2. Validate the changes with a `commit`

If you use the command line, this will give:

```
git add .
git commit -m "add cat emoji"
```

4️⃣ Make a **second commit** to add a koala emoji (:koala:) then push the local changes: + In `VSCode`, click on the `Publish Branch` button. + Otherwise, if you use the command line, you will have to type `git push origin issue-1`

5️⃣ In `Github`, you should see

> `issue-1 had recent pushes XX minutes ago`.

Click on `Compare & Pull Request`. Give your *pull request* an informative title. In the message below, type

> `- close #1`

The dash is a little trick so that `Github` replaces the issue number with its title.

Click on `Create Pull Request` but **do not validate the merge**, we will do it in a second step.

Having put a `close` message followed by an issue number `#1` will automatically close *issue 1* when you do the *merge*. In the meantime, you have created a link between the *issue* and the *pull request*

While you are at it, you can add the `Markdown` label on the right.

6️⃣ Locally, go back to `main`. In the `Jupyter` interface, just click on `main` in the list of branches. In `VSCode`, the list of branches appears when you click on the name of the current branch (`issue-1` in theory at this stage). If you are on the command line, you need to run

```
git checkout main
```

`checkout` is a `Git` command that lets you navigate from one branch to another (or even from one commit to another).

Add a sentence after your text in the `README.md` (do not touch the title!). You can notice that the emojis are not in the title, this is normal, you have not merged the versions yet

7️⃣ Make a commit and a push. On the command line, this gives

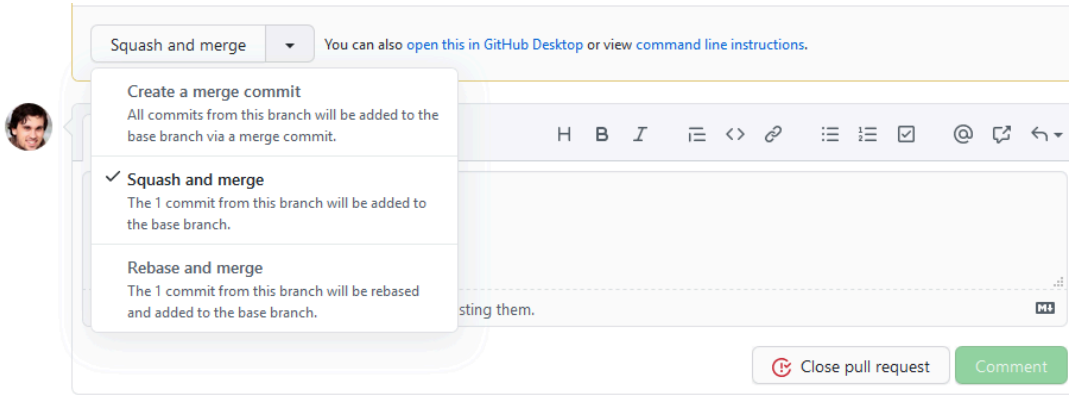
```
git add .
git commit -m "add a third line"
git push origin main
```

8️⃣ On `Github`, click on `Insights` at the top of the repository then, on the left, on `Network` (this is only possible if you made your repository public).

You should see the tree structure of your repository appear. You can see `issue-1` as a ramification and `main` as the trunk.

The goal is now to bring the changes made in `issue-1` into the main branch. Go back to the `Pull Requests` tab. There, change the type of `merge` to `Squash and Merge`, as below (a small piece of advice: always choose this *merge* method).

Once this is done, you can go back to `Insights` then `Network` to check that everything went as planned.



9️⃣ Delete the branch (*branch > delete this branch*). Since it is merged, it will no longer be needed. Keeping it risks leading to unintended `push`es on it.

Tip

The *Squash and Merge* option combines all the commits of a branch (potentially very numerous) into a single one in the target branch. On large projects, this avoids branches with thousands of *commits*.

I recommend always using this technique and not the others. To disable the other techniques, you can go to `Settings` and, in the `Merge button` section, keep only the `Allow squash merging` method checked

We now have all the basics needed to move on to the next step in our `Git` journey: collaborative work. Nevertheless, `Git` should not be used only in collective projects. Even when working alone, the quality gains from using `Git` are unmatched.

5 Additional case studies

The previous exercises follow a guided path on a real `Github` repository. The case studies below are complementary: they run directly on this web page[5], reproducing the behaviour of a command line. This way, you can experiment without fear: the `Undo` button cancels the last command, and `Reset` brings the exercise back to its initial state.

The expected steps are ticked automatically when you reach the expected state.

The bottom panel is updated after each command: `Files` (working directory, *staging area*, repository), `Changes` (what has changed since the last *commit*) and `History` (the graph of *commits*).

5.1 Case study 1: `git add` freezes a version, not a file

In exercise 5, we added files to the *staging area* and then made a *commit* without touching the file in between. But what happens if we modify the file again **after** staging it?

> Case study 1: modifying a file after git add
>
> We are in a situation where a file (`analysis.py`) is already tracked by `Git` and versioned.
>
> 1. Add a line to `analysis.py` then stage the file with `git add`.
> 2. Add a second line **without** staging again, then run `git status`. What do you notice? Also look at the `Files` panel.
> 3. Make a *commit* then run `git status` again.
> 4. Finish by committing the second change.
>
> *Try this case study with the interface below.*

This exercise is interactive. Open the HTML version of this lesson to work through it in a browser.

- Stage a change then modify the same file again
- Commit and notice that the file is not “clean”
- Commit the second change

The file appears in two categories at the same time:

1. `Changes to be committed` (the version staged with `git add`)
2. `Changes not staged for commit` (the more recent version, in the working directory).

The *commit* only records the first one. This mechanism explains why we run `git status` before making a *commit*: it avoids forgetting changes, or sending changes we thought we had validated.

5.2 Case study 2: abandoning an experiment

In exercise 9, the `issue-1` branch is merged into `main`. But a branch is also made to **try** things out: if the experiment leads nowhere, you abandon the branch and `main` was never touched.

[5] These exercises use the `quarto-git-sandbox` extension. The terminal recognises a subset of `Git` commands (type `help` to see them) or `Linux` commands such as `echo` or `cat`.

Case study 2: a disposable branch

You want to test a `LightGBM` model without the risk of destabilising `main`.

1. Create the branch `lightgbm-trial` and switch to it (`git checkout -b lightgbm-trial`).
2. Create a file `lightgbm.py` and make two *commits* on this branch.
3. Go back to `main` with `git checkout main` and check with `ls` that `lightgbm.py` no longer exists.
4. The experiment is not conclusive: delete the branch with `git branch -D lightgbm-trial`. Look at the `History` panel.

Try this case study with the interface below.

This exercise is interactive. Open the HTML version of this lesson to work through it in a browser.

- Create the branch lightgbm-trial and switch to it
- Create lightgbm.py on it and make two commits
- Go back to main, where lightgbm.py does not exist
- Delete the branch with git branch -D; main stays at two commits

5.3 Case study 3: an urgent fix in the middle of a work in progress

Here is a frequent case, even when working alone: you are in the middle of a work branch and you notice a problem that needs to be fixed right away on `main`. The two lifelines diverge, and then you bring them back together.

Case study 3: fixing main along the way

You are on the `plot` branch, which is half finished. You notice a typo in the title of the `README.md` (“forcast” instead of “forecast”).

1. Switch to `main` (`git checkout main`), fix the title of the `README.md` and make a *commit*.
2. Go back to `plot` and add a *commit* to keep the work going.
3. Return to `main` and merge the branch with `git merge plot`.
4. Look at the `History` panel: what does the hollow circle represent?

Try this case study with the interface below.

This exercise is interactive. Open the HTML version of this lesson to work through it in a browser.

- On main, fix the title of the README.md and commit
- Go back to plot and make a new commit on it
- Merge plot into main (a merge commit appears)

To compare with exercise 9: here the merge is done locally and **keeps** all the *commits* of the branch, plus the merge *commit*. This is what the `Squash and merge` option of `Github` replaces: it condenses everything into a single *commit* and gives a more readable history.

Informations additionnelles

Python environment

This site was built automatically through a `Github` action using the `Quarto` reproducible publishing software (version 1.10.18).

The environment used to obtain the results is reproducible via uv. The `pyproject.toml` file used to build this environment is available on the `linogaliana/python-datascientist` repository

```
[project]
name = "python-datascientist"
version = "0.1.0"
description = "Source code for Lino Galiana's Python for data science course"
readme = "README.md"
requires-python = "≥3.13,<3.14"
dependencies = [
    "altair≥6.0.0",
    "cartiflette≥0.2",
    "contextily==1.6.2",
    "duckdb≥0.10.1",
    "folium≥0.19.6",
    "gdal==3.11.4",
    "graphviz==0.20.3",
    "great-tables≥0.12.0",
    "gt-extras≥0.0.8",
    "ipykernel≥6.29.5",
    "jupyter≥1.1.1",
    "jupyter-cache≥1.0.0",
    "kaleido≥0.2.1",
    "langchain-community≥0.3.27",
    "loguru==0.7.3",
    "markdown≥3.8",
    "nbclient≥0.10.0",
    "nbformat≥5.10.4",
    "nltk≥3.9.1",
    "openpyxl≥3.1.5",
    "pandas≥3.0",
    "pip≥25.1.1",
    "plotly≥6.1.2",
    "plotnine≥0.15",
    "polars≥1.8.2",
    "py7zr≥1.1.3",
    "pyarrow≥17.0.0",
    "pynsee≥0.1.8",
    "python-dotenv≥1.0.1",
    "python-frontmatter≥1.1.0",
    "pywaffle≥1.1.1",
    "requests≥2.32.3",
    "scikit-image≥0.24.0",
    "scikit-learn≥1.8.0",
    "scipy≥1.13.0",
    "seaborn≥0.13.2",
    "selenium<4.39.0",
    "spacy≥3.8.4",
    "webdriver-manager≥4.0.2",
    "wordcloud==1.9.3",
]

[tool.uv.sources]
gdal = [
  { index = "gdal-wheels", marker = "sys_platform == 'linux'" },
  { index = "geospatial_wheels", marker = "sys_platform == 'win32'" },
]
```

```
[[tool.uv.index]]
name = "geospatial_wheels"
url = "https://nathanjmcdougall.github.io/geospatial-wheels-index/"
explicit = true

[[tool.uv.index]]
name = "gdal-wheels"
url = "https://gitlab.com/api/v4/projects/61637378/packages/pypi/simple"
explicit = true

[dependency-groups]
dev = [
    "nb-clean>=4.0.1",
]
```

To use exactly the same environment (version of `Python` and *packages*), please refer to the documentation for `uv`.

Bibliography

[1] Arcep, “L'empreinte carbone du numérique,” technical report, 2019.